AP PRECALCULUS

Unit 1 • Lesson 1.7 • Notes (filled in)

Topic 1.7 • Rational Functions and End Behavior

Notes (filled in)

Learning goals

- Determine the end behavior of a rational function by comparing the degrees of numerator and denominator.
- Find horizontal and slant asymptotes, and write end behavior with limits.

Vocabulary. rational function, horizontal asymptote, slant asymptote, quotient of leading terms.

1. Quotient of leading terms

For $r(x) = \dfrac{p(x)}{q(x)}$, as $x \to \pm\infty$ the function behaves like $\dfrac{\text{leading term of } p}{\text{leading term of } q}$.

Key idea

$\deg p < \deg q$: horizontal asymptote $y = 0$

$\deg p = \deg q$: horizontal asymptote $y =$ ratio of the leading coefficients

$\deg p = \deg q + 1$: slant asymptote, found by polynomial division

$\deg p > \deg q + 1$: behaves like a polynomial of degree $\deg p - \deg q$

Example 1 • Limits at infinity

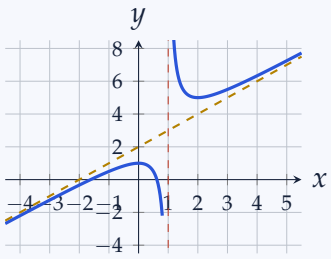
Find $\lim_{x\to\infty}$ for $\dfrac{3x+1}{x^2-4}$, $\dfrac{8x^2-x}{2x^2+5}$, $\dfrac{x^3}{x^2+1}$.

0 (degree of the denominator is larger). $\frac{8}{2} = 4$. Behaves like x: ∞ (and as $x \to -\infty$, $-\infty$).

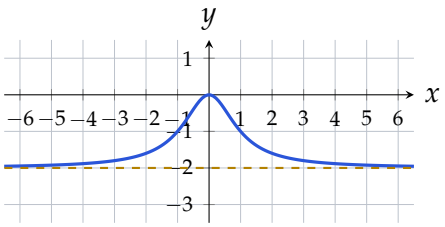
Example 2 • Slant asymptote

Find the slant asymptote of $s(x) = \dfrac{x^2+x-1}{x-1}$.

$x^2+x-1 = (x-1)(x+2)+1$, so $s(x) = x+2+\dfrac{1}{x-1}$. As $x \to \pm\infty$, $\frac{1}{x-1} \to 0$: slant asymptote $y = x+2$.



Example 3 • From a graph



The graph of $r(x) = \frac{-2x^2}{x^2+1}$ is shown. Write the end behavior with limits.

$\lim_{x\to\infty} r(x) = -2$ and $\lim_{x\to-\infty} r(x) = -2$ (horizontal asymptote $y = -2$).

Common mistakes

- Reading the asymptote from the constant terms.
- Believing a graph can never cross a horizontal asymptote: it can, for moderate x.

Check your understanding

1. Horizontal asymptote of $\dfrac{5-2x^3}{x^3+x}$: $y = -2$.
2. Slant asymptote of $\dfrac{2x^2+3x}{x+1}$: $y = 2x+1$.